

3751. Total Waviness of Numbers in Range I

Solved ●

Medium  Topics  Hint

You are given two integers `num1` and `num2` representing an **inclusive** range `[num1, num2]`.

The **waviness** of a number is defined as the total count of its **peaks** and **valleys**:

- A digit is a **peak** if it is **strictly greater** than both of its immediate neighbors.
- A digit is a **valley** if it is **strictly less** than both of its immediate neighbors.
- The first and last digits of a number **cannot** be peaks or valleys.
- Any number with fewer than 3 digits has a waviness of 0.

Return the total sum of waviness for all numbers in the range `[num1, num2]`.

Example 1:

Input: `num1 = 120, num2 = 130`

Output: 3

Explanation:

In the range `[120, 130]`:

- `120`: middle digit 2 is a peak, waviness = 1.
- `121`: middle digit 2 is a peak, waviness = 1.
- `130`: middle digit 3 is a peak, waviness = 1.
- All other numbers in the range have a waviness of 0.

Thus, total waviness is `1 + 1 + 1 = 3`.

Example 2:

Input: `num1 = 198, num2 = 202`

Output: 3

Explanation:

In the range `[198, 202]`:

- `198`: middle digit 9 is a peak, waviness = 1.
- `201`: middle digit 0 is a valley, waviness = 1.
- `202`: middle digit 0 is a valley, waviness = 1.
- All other numbers in the range have a waviness of 0.

Thus, total waviness is `1 + 1 + 1 = 3`.

Example 3:

Input: `num1 = 4848, num2 = 4848`

Output: 2

Explanation:

Number `4848`: the second digit 8 is a peak, and the third digit 4 is a valley, giving a waviness of 2.

Constraints:

- `1 <= num1 <= num2 <= 105`

Seen this question in a real interview before? 1/6

Yes No

Accepted 139,328/159.2K | Acceptance Rate 87.5%

Topics



Hint 1



Discussion (145)



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